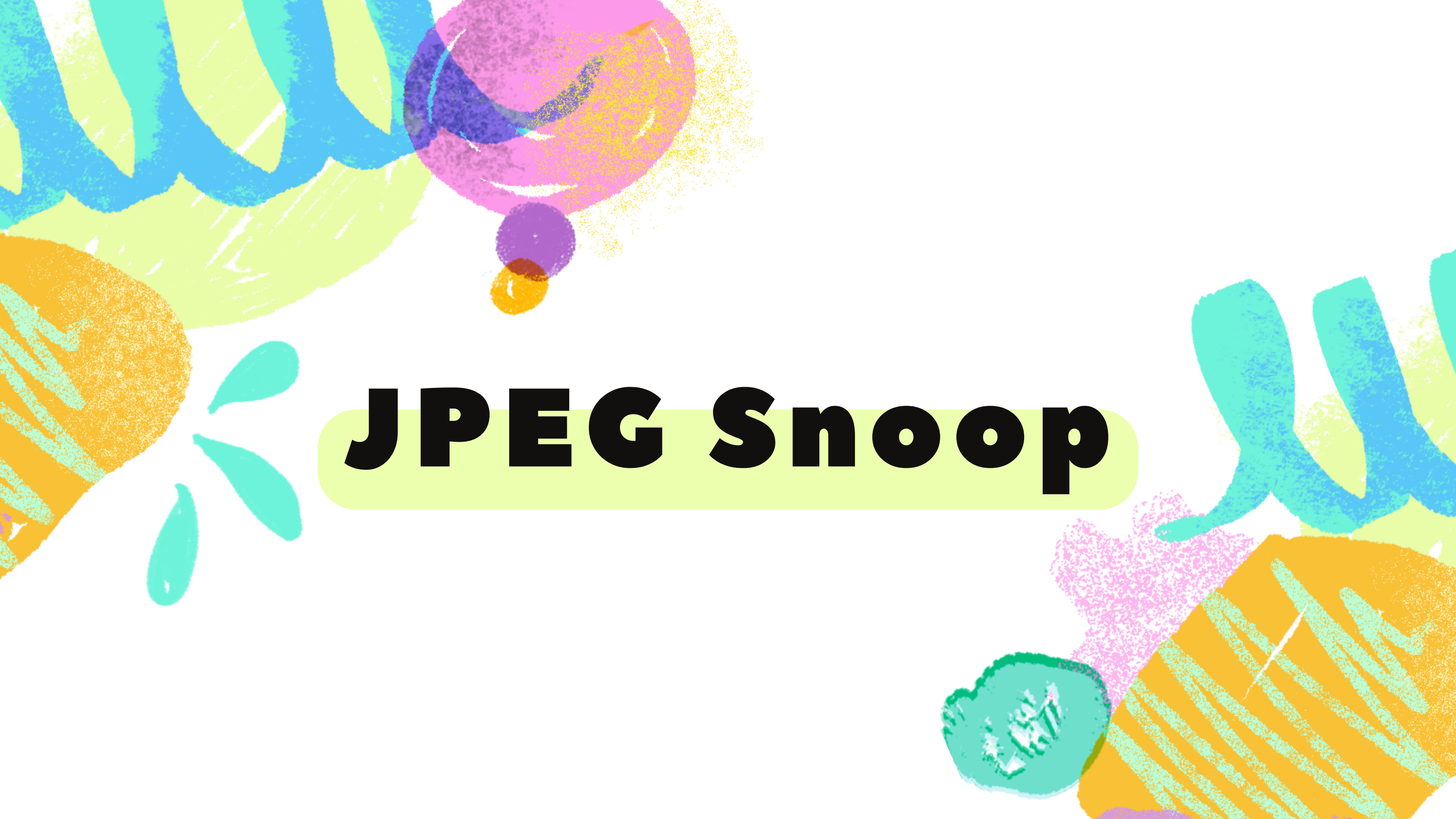


The background is a vibrant, abstract collage of various colors and textures. It includes large, soft-edged shapes in shades of yellow, pink, blue, and orange. There are also smaller, more defined elements like a blue shape with yellow dots, a green shape with white dots, and a yellow shape with orange dots. The overall style is reminiscent of a child's drawing or a modern, playful graphic design.

JPEG

SNOOP

**ARTUR, MARINA &
PEDRO**



JPEG Snoop

Licença

JPEGsnoop é gratuito para uso pessoal e comercial de acordo com a Licença Pública Geral GNU versão 2.0 (GPLv2).



Sistemas Suportados

Windows (MFC) **Visual Studio**

Compilação no Visual Studio (Visual C++)

05



Funcionalidades

1. Decodificação de imagens JPEG, AVI (MJPG), THM, DNG, CRW, CR2, NEF, ORF, PEF, RAW, MOV e PDF;
2. Processar batches, salvar e exportar as informações coletadas
3. Análise de MCU com decodificação detalhada;
4. Extração de imagens embedded JPEG;
5. Detecção de imagens editadas através de análise de assinatura de compressão;
6. Relato de metadados da imagem (EXIF);
7. Processamento em lote de arquivos;
8. Recuperação de dados de imagem JPEG corrompidos;
9. Não é preciso instalação!

Authentication of Digital Image using Exif Metadata and Decoding Properties

D. P. Gangwar, Anju Pathania

Central Forensic Science Laboratory, Sector 36-A, Chandigarh, DFSS, MHA, Government of India

- Pesquisa com intuito de verificar a autenticidade de imagens através de dados obtidos por várias ferramentas, dentre elas, JPEGsnoop.
- Feita inteiramente com ferramentas OSINT (Open Source Intelligence)

https://www.researchgate.net/profile/D-P-Gangwar/publication/343650073_AUTHENTICATION_OF_DIGITAL_AUDIO_RECORDING_USING_FILE'S_SIGNATURE_AND_METADATA_PROPERTIES/links/60d740eca6fdccb745e9054a/AUTHENTICATION-OF-DIGITAL-AUDIO-RECORDING-USING-FILES-SIGNATURE-AND-METADATA-PROPERTIES.pdf

Já nas imagens editadas, o marcador APP0

```
Filename: [C:\Users\dp\Desktop\20180808_091254.jpg]
Filesize: [2960340] Bytes

Start Offset: 0x00000000
*** Marker: SOI (xFFD8) ***
Offset: 0x00000000

*** Marker: APP1 (xFFE1) ***
Offset: 0x00000002
Length      = 28738
Identifier  = [Exif]
Identifier TIFF = 0x[49492A00 08000000]
Endian      = Intel (little)
TAG Mark x002A = 0x002A

EXIF IFD0 @ Absolute 0x00000014
Dir Length = 0x000D
[Make      ] = "samsung"
[Model     ] = "SM-G600FY"
[Orientation] = 6 = Row 0: right, Col
[XResolution] = 72/1
[YResolution] = 72/1
[ResolutionUnit] = Inch
[Software  ] = "G600FYDDU1BRD2"
```

Filesize: [2421979] Bytes

Start Offset: 0x00000000

*** Marker: SOI (xFFD8) ***

Offset: 0x00000000

*** Marker: APP0 (xFFE0) ***

Offset: 0x00000002

Length = 16

Identifier = [JFIF]

version = [1.1]

density = 96 x 96 DPI (dots per inch)

thumbnail = 0 x 0

*** Marker: APP1 (xFFE1) ***

Offset: 0x00000014

Length = 3226

Identifier = [Exif]

Identifier TIFF = 0x[4D4D002A 00000008]

Endian = Motorola (big)

TAG Mark x002A = 0x002A

EXIF IFD0 @ Absolute 0x00000026

Dir Length = 0x000C

[Make] = "samsung"

[Model] = "SM-G600FY"

[XResolution] = 72/1

[YResolution] = 72/1

[ResolutionUnit] = Inch

[Software] = "PhotoScape"

[DateTime] = "2018:08:08 09:12:54"

[YCbCrPositioning] = Centered

[ExifOffset] = @ 0x00E0

[GPSOffset] = @ 0x0C80

Offset to Next IFD = 0x00000000

Nas imagens originais, foi possível
perceber o marcador APP1



Demonstração

Almoço em família.jpg



*** Searching Compression Signatures ***

Signature: 01603E81833C1DF345A9ED9926E47A12
Signature (Rotated): 0135ED76EC3CE185F34E650FE624DC7F
File Offset: 0 bytes
Chroma subsampling: 2x1
EXIF Make/Model: OK [Nokia] [6275]
EXIF Makernotes: NONE
EXIF Software: OK [V BL105V0100.nep]

Searching Compression Signatures: (3347 built-in, 1 user(*))

EXIF.Make / Software	EXIF.Model	Quality	Subsamp Match?
*CAM:[Nokia	] [6275	] [	] Yes

Based on the analysis of compression characteristics and EXIF metadata:

ASSESSMENT: Class 3 - Image has high probability of being original
Note that EXIF Software field is set (typically contains Firmware version)

Start Offset: 0x00000000
*** Marker: SOI (xFFD8) ***
Offset: 0x00000000

*** Marker: APP1 (xFFE1) ***
Offset: 0x00000002
Length = 2078
Identifier = [Exif]
Identifier TIFF = 0x[4D4D002A 00000008]
Endian = Motorola (big)
TAG Mark x002A = 0x002A

EXIF IFD0 @ Absolute 0x00000014
Dir Length = 0x0009
[Make] = "Nokia"
[Model] = "6275"
[Orientation] = 1 = Row 0: top, Col 0: left
[XResolution] = 72/1
[YResolution] = 72/1
[ResolutionUnit] = Inch
[Software] = "V BL105V0100.nep"
[YCbCrPositioning] = Centered
[ExifOffset] = @ 0x00A6
Offset to Next IFD = 0x000000F4

EXIF IFD1 @ Absolute 0x00000100
Dir Length = 0x0006
[Compression] = JPEG
[XResolution] = 72/1
[YResolution] = 72/1
[ResolutionUnit] = Inch
[JpegIFOffset] = @ +0x0152 = @ 0x015E
[JpegIFByteCount] = 0x[000006C4] / 1732
Offset to Next IFD = 0x00000000

EXIF SubIFD @ Absolute 0x000000B2
Dir Length = 0x0006
[ExifVersion] = 02.20
[ComponentsConfiguration] = [Y Cb Cr .]
[FlashPixVersion] = 01.00
[ColorSpace] = sRGB
[ExifImageWidth] = 1600
[ExifImageHeight] = 1200

*** Marker: APP2 (xFFE2) ***
Offset: 0x00000822
Length = 13
Identifier = []
Not supported. Skipping remainder.

*** Marker: DQT (xFFDB) ***
Define a Quantization Table.
Offset: 0x00000831
Table length = 132

Precision=8 bits
Destination ID=0 (Luminance)
DOT. Row #0: 6 4 4 6 9 16 20 24